

Sephora Reviews Analytics Pipeline

Raw CSVs -> clean data -> 3NF OLTP -> star schema -> Airflow -> live Streamlit

1.09M

warehouse facts

16

Airflow tasks

10

analytics views

51

passing tests

Source data first; rules second

The catalogue and review stream arrive at different grains, so profiling drives the cleaning contract.

1,094,411

raw reviews

8,494

catalogue products

503,216

review authors

304

brands

14

profiling checks

Deduplicate at the right grain

Remove 1,040 rows on (author, product, date). Keep 4,485 legitimate re-reviews that a coarser key would destroy.

Standardize, do not invent

Grey -> gray. notSureST -> Unknown at the staging boundary. Helpfulness nulls stay undefined when nobody voted.

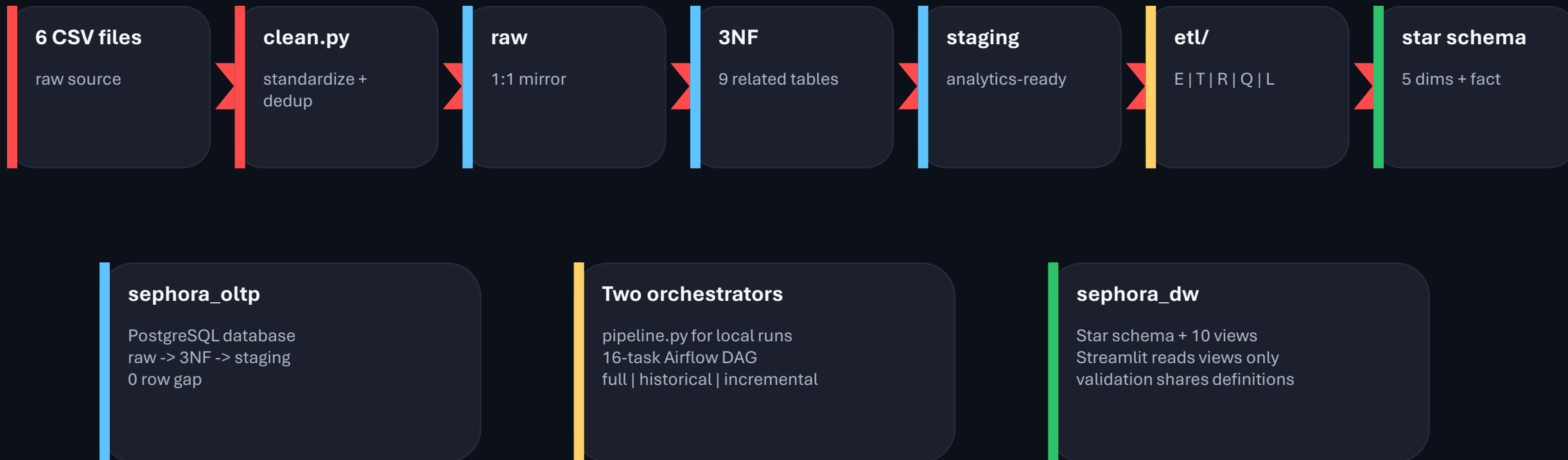
Keep traceability

Cleaning drops rows, never columns. Every source column reaches raw; scope trims happen explicitly at raw -> 3NF.

Result: 1,093,371 clean reviews - exactly the population reconciled through 3NF, staging, and the warehouse.

A traceable path from files to decisions

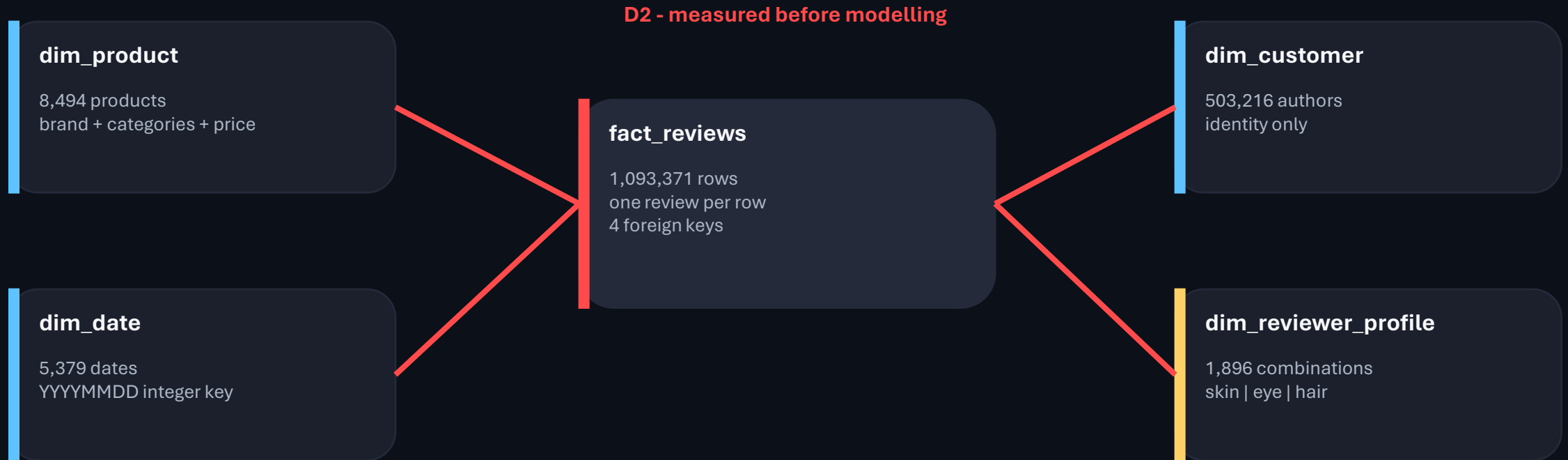
Each layer has one job; the same Python package runs locally or under Airflow.



Grain of fact_reviews: one row per review. PostgreSQL generates every surrogate key.

The profile belongs to the review

A one-row-per-author customer dimension would silently assign the wrong skin profile to roughly one review in seven.



22,503 authors changed profile attributes
149,788 reviews affected | 13.69% of the dataset

Make silent data loss impossible

The course reference structure is preserved, with reconciliation and severity-aware gates added where this dataset needs them.



Hard failures stop before writes | warnings remain visible | ON CONFLICT DO NOTHING enforces idempotency

The DAG ran both paths successfully

Controlled verification on 12 Aug 2026 | 16 tasks | 15 success + the intentionally skipped one_failed watcher

verification_historical_20260812 Success

Run Type: Manual | Start Date: 2026-08-12 19:12:03 | End Date: 2026-08-12 19:14:46 | Duration: 00:02:43 | Triggering User: airflow | Dag Version(s): v3

Task Instances

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
watch_for_failure		Skipped	2026-08-12 19:14:46	0	@task	
cleanup_staging		Success	2026-08-12 19:14:42	1	@task	00:00:03.686
load_fact_from_staging		Success	2026-08-12 19:14:01	1	@task	00:00:40.355
quality_check_fact_staged		Success	2026-08-12 19:13:55	1	@task	00:00:06.480
transform_fact_staged		Success	2026-08-12 19:13:05	1	@task	00:00:48.407
extract_fact_to_staging		Success	2026-08-12 19:12:15	1	@task	00:00:43.902
load_date_dimension		Success	2026-08-12 19:12:15	1	@task	00:00:01.159
load_product_from_staging		Success	2026-08-12 19:12:20	1	@task	00:00:02.235
extract_product_to_staging		Success	2026-08-12 19:12:18	1	@task	00:00:01.216
load_reviewer_profile_from_staging		Success	2026-08-12 19:12:17	1	@task	00:00:00.967
extract_reviewer_profile_to_staging		Success	2026-08-12 19:12:15	1	@task	00:00:01.411
load_customer_from_staging		Success	2026-08-12 19:12:21	1	@task	00:00:05.459
extract_customer_to_staging		Success	2026-08-12 19:12:15	1	@task	00:00:06.197
load_brand_from_staging		Success	2026-08-12 19:12:17	1	@task	00:00:00.880
extract_brand_to_staging		Success	2026-08-12 19:12:15	1	@task	00:00:01.124

Historical re-offer | **SUCCESS** | 164 seconds

verification_incremental_20260812 Success

Run Type: Manual | Start Date: 2026-08-12 19:15:23 | End Date: 2026-08-12 19:15:50 | Duration: 00:00:27.228 | Triggering User: airflow | Dag Version(s): v3

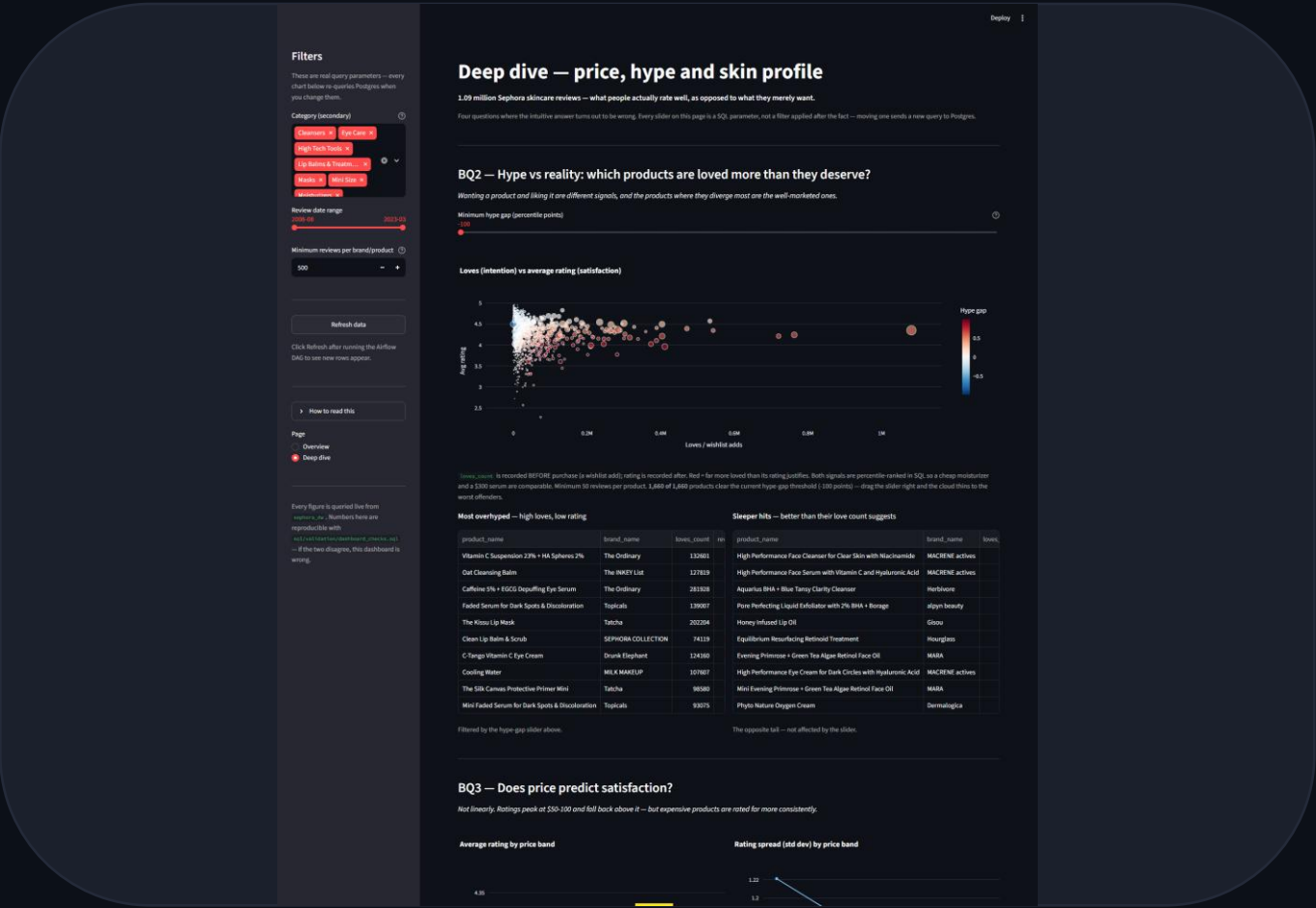
Task Instances

Task ID	Map Index	State	Start Date	Try Number	Operator	Duration
watch_for_failure		Skipped	2026-08-12 19:15:49	0	@task	
cleanup_staging		Success	2026-08-12 19:15:48	1	@task	00:00:00.887
load_fact_from_staging		Success	2026-08-12 19:15:42	1	@task	00:00:04.483
quality_check_fact_staged		Success	2026-08-12 19:15:41	1	@task	00:00:00.958
transform_fact_staged		Success	2026-08-12 19:15:37	1	@task	00:00:03.191
extract_fact_to_staging		Success	2026-08-12 19:15:25	1	@task	00:00:03.438
load_date_dimension		Success	2026-08-12 19:15:25	1	@task	00:00:00.881
load_product_from_staging		Success	2026-08-12 19:15:30	1	@task	00:00:00.990
extract_product_to_staging		Success	2026-08-12 19:15:28	1	@task	00:00:01.261
load_reviewer_profile_from_staging		Success	2026-08-12 19:15:28	1	@task	00:00:00.907
extract_reviewer_profile_to_staging		Success	2026-08-12 19:15:25	1	@task	00:00:01.138
load_customer_from_staging		Success	2026-08-12 19:15:31	1	@task	00:00:04.285
extract_customer_to_staging		Success	2026-08-12 19:15:25	1	@task	00:00:04.953
load_brand_from_staging		Success	2026-08-12 19:15:27	1	@task	00:00:00.791
extract_brand_to_staging		Success	2026-08-12 19:15:25	1	@task	00:00:00.840

Incremental restore | **SUCCESS** | 27 seconds

Deleted only the verified 2023 fact slice -> restored all 49,503 rows -> watermark returned to 2023-03-21.

The dashboard leads with decisions, not decoration



Price vs satisfaction: an inverted U



Ratings peak at \$50-100, then fall back above \$100. Variation narrows as price rises.

- Hype separates wishlist intention from satisfaction
- March 2023 is labelled partial
- Skin-profile effects are real but weak
- Every slider binds into SQL

Default page: summary first. Deep dive: hype, price, reviewer profile, and review-length polarisation.

A complete, reproducible capstone

The final state is measured, source-backed, and runnable from an empty Docker environment.

1,093,371

facts = staging rows

0

orphan fact keys

0

duplicate fact keys

51 + 11

host tests + DAG asserts

Live demo in four moves

1. Show the successful incremental run
2. Open the live Overview KPIs
3. Move one Deep dive SQL control
4. Reconcile views to fact_reviews

Where to verify it

README.md - setup + diagrams
docs/ - decisions + evidence
dashboard/ - app + metric model
sql/validation/ - source-of-truth checks

Questions?